Center Estates ERP

CONSTRUCTION OVERSIGHT & FIELD TELEMETRY

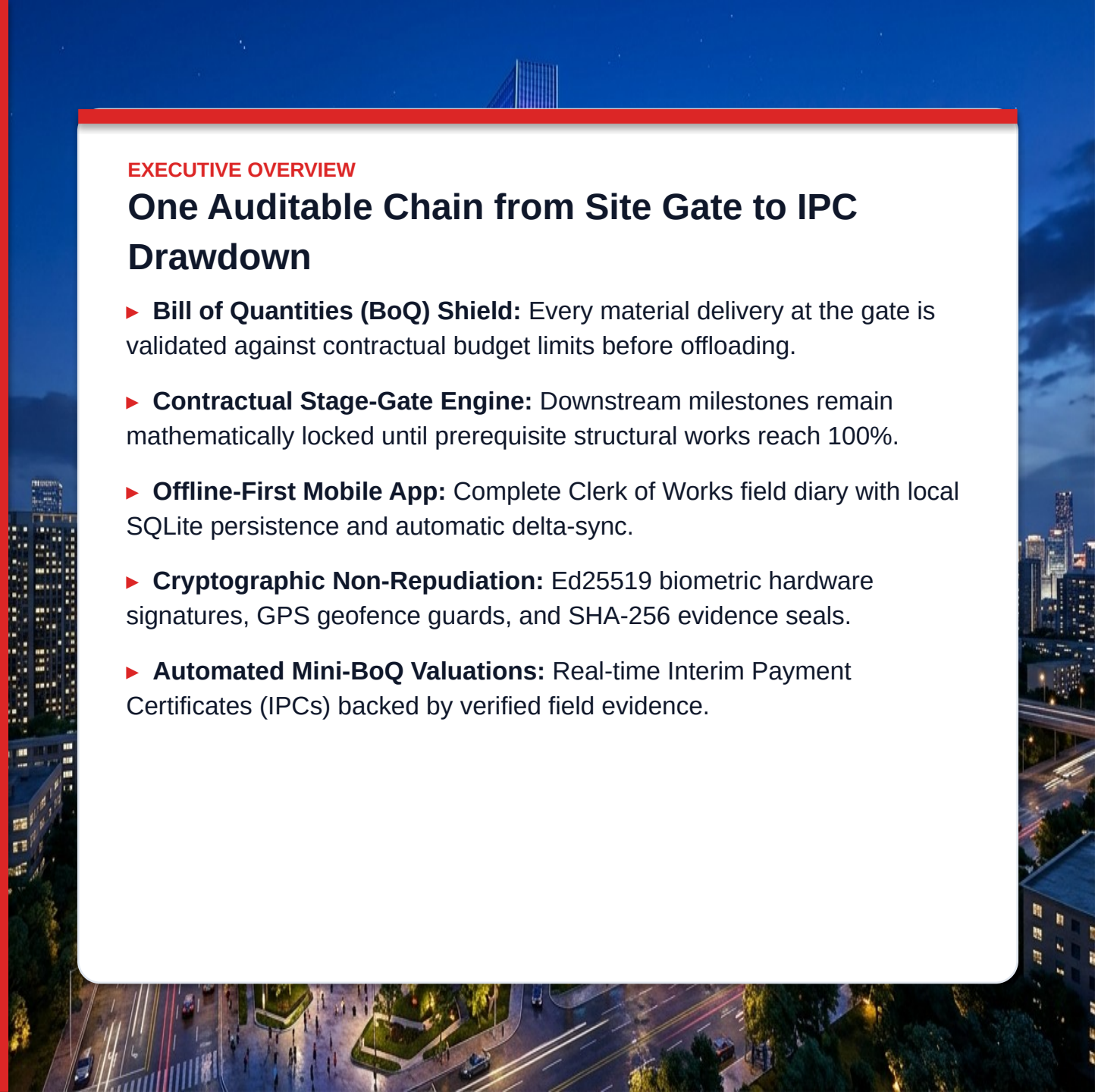
LIVE STAKEHOLDER BRIEFING

Eliminating Capital Leakage in Real Estate Construction.

Connecting the physical site gate directly to bank drawdown approvals via BoQ Allowance Shields, Contractual Stage-Gates, and Offline-First Mobile Telemetry.

100%	SHA-256	0% Leakage
BoQ-Bound Gate Intake	Tamper-Evident Seals	Stage-Gate Protected

Featured Project: Karam Apartments — Lang'ata (PRJ-KRM-01) • centerestates.com



EXECUTIVE OVERVIEW

One Auditable Chain from Site Gate to IPC Drawdown

- **Bill of Quantities (BoQ) Shield:** Every material delivery at the gate is validated against contractual budget limits before offloading.
- **Contractual Stage-Gate Engine:** Downstream milestones remain mathematically locked until prerequisite structural works reach 100%.
- **Offline-First Mobile App:** Complete Clerk of Works field diary with local SQLite persistence and automatic delta-sync.
- **Cryptographic Non-Repudiation:** Ed25519 biometric hardware signatures, GPS geofence guards, and SHA-256 evidence seals.
- **Automated Mini-BoQ Valuations:** Real-time Interim Payment Certificates (IPCs) backed by verified field evidence.

01 • INDUSTRY CHALLENGE

The Three Critical Blind Spots in Construction Execution

Why 15% to 25% of commercial development capital is lost between the site gate and the finance office.

01 Phantom & Over-Deliveries

Paper delivery notes signed at the site gate with zero live cross-reference to the approved Bill of Quantities.

- **Budget Overruns:** Materials delivered beyond contractual allowances without real-time visibility.
- **Supplier Opacity:** No standardized tracking of batch quality or approved manufacturers.
- **Retrospective Discovery:** Variances only uncovered months later during final account audits.

02 Premature Progress Claims

Contractors claiming completion on downstream finishes before critical structural foundations are certified.

- **Front-Loaded Claims:** Capital disbursed ahead of actual physical progress on site.
- **Subjective Valuations:** Visual estimates disconnected from measured BoQ sub-elements.
- **Dispute Exposure:** Lack of immutable stage-gate prerequisites triggers arbitration.

03 Disconnected Site Diaries

Clerk of Works observations scattered across paper logbooks, email threads, and unverified WhatsApp photos.

- **Unverified Media:** Site photos lack hardware GPS coordinates and tamper-evident hashes.
- **Missed Concrete Tests:** 7, 14, and 28-day cylinder crushing tests forgotten or unlinked.
- **Snag Amnesia:** Structural defects logged verbally fail to reach the QS before payment.

GOVERNANCE GUARANTEE **Traditional oversight relies on trust and post-mortem audits. Center Estates enforces proactive digital stage-gates.**

02 • UNIFIED PLATFORM ARCHITECTURE

End-to-End Architecture: Mobile Field App ↔ REST API ↔ Web ERP

A single continuous operating record connecting field engineers, quantity surveyors, and project financiers.

APP Mobile Field App (Expo / RN)

Authoritative source of physical site truth for Clerks of Works and Resident Engineers.

- **Offline-First SQLite:** TinyBase reactive store persists all logs & drafts without 4G.
- **Rugged Field UX:** Searchable BoQ picker and one-tap Kenyan supplier chips.
- **GPS Geofencing:** Hardware coordinate lock against project site boundaries.
- **Biometric Sign-Off:** Ed25519 hardware key signatures on every submission.

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API Central API (Django & MySQL)

Production governance engine running at api.centerestates.com.

- **Atomic Delta-Sync:** Idempotent /sync/pull and /sync/push endpoints.
- **Stage-Gate Validator:** Rejects progress >0% if prerequisite milestones <100%.
- **SHA-256 Evidence Seals:** Cryptographic hash computed over payload + photo bytes.
- **Role-Based Access:** Strict separation of Clerk, Consultant, and Owner permissions.

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WEB Office Web ERP (Next.js 14)

Executive command center for Developers, Project Managers, and QSs.

- **CPM Work Plan & Gantt:** Predecessor scheduling and critical path tracking.
- **Bill of Quantities (BoQ):** Live line-item budgets, rates, and variance shields.
- **Mini-BoQ Valuations:** Automated Interim Payment Certificates (IPCs).
- **Live Snag Kanban:** Instant visibility into field-reported quality defects.

GOVERNANCE GUARANTEE **Immutable Field Truth: Office users can review and certify, but can never alter or delete raw field evidence.**

03 • FIELD MATERIAL CONTROL

Site Gate Deliveries Bound to BoQ Allowances

Preventing material over-delivery and budget leakage before the truck leaves the site perimeter.

01 Searchable BoQ Item Picker

Clerk selects directly from the project's approved Bill of Quantities (e.g. BOQ-02.04 Cement). Automatically binds item code, unit, and contractual rate.

02 Kenyan Supplier Quick-Chips

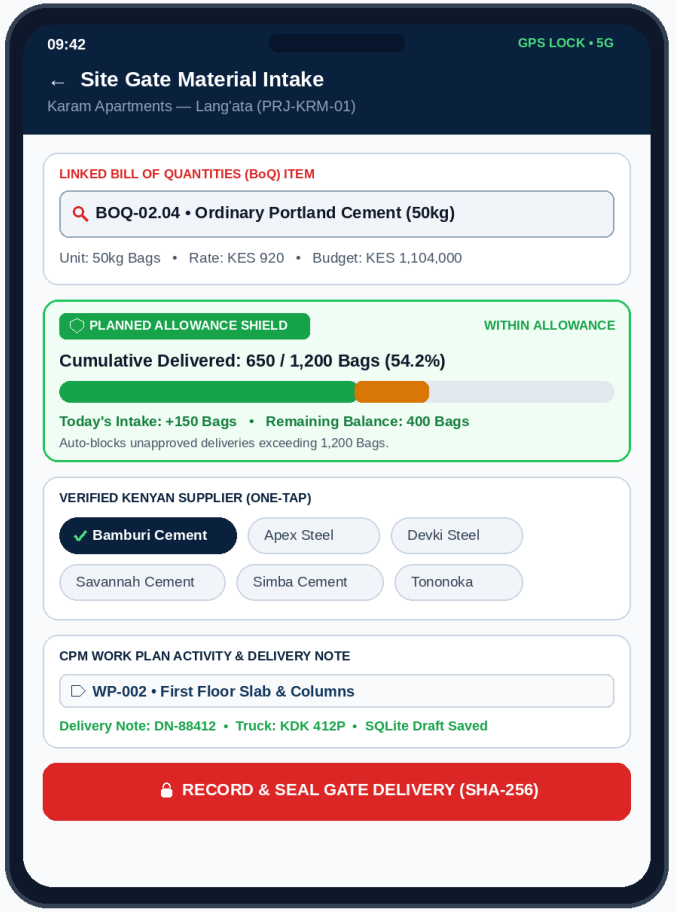
One-tap verified supplier chips (Bamburi, Apex Steel, Devki, Savannah, Simba, Tononoka) eliminate typos and standardize procurement audits.

03 Planned Allowance Shield

Live progress bar compares cumulative delivered quantity against total BoQ allowance. Triggers an immediate red warning if intake exceeds budget.

04 Work Plan Task Tagging

Every material intake is tagged to a specific CPM Work Plan activity (e.g. WP-002 First Floor Slab) and persists locally via useFormDraft.



GOVERNANCE GUARANTEE — **100% Traceability: Every bag of cement and ton of steel is linked to a BoQ line item, supplier, and work plan task.**

04 • CONTRACTUAL GOVERNANCE

The Stage-Gate Engine: Mathematical Payout Protection

Downstream milestones cannot advance above 0% until all prerequisite structural works reach 100%.

MS-01: Substructure & Raft
50% IN PROGRESS
Prerequisite Incomplete

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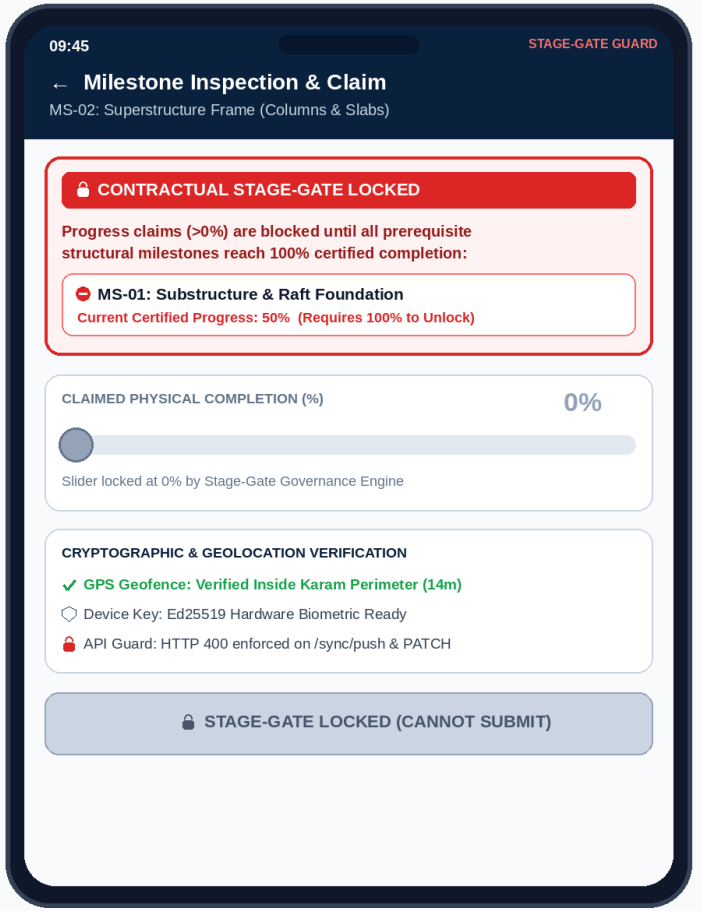
MS-02: Superstructure Frame
🔒 LOCKED AT 0%
Blocked by MS-01 (<100%)

→

MS-03: Roofing & Finishes
🔒 LOCKED AT 0%
Blocked by MS-02 (<100%)

🔒 Dual-Layer Enforcement (Mobile UI + Backend API)

- **Mobile Stage-Gate Lock Banner:** MilestoneDetailScreen evaluates evaluateMilestoneGovernance() in real time, displaying the exact blocking predecessors and disabling the submit button.
- **Strict Backend Validation:** Both PATCH /milestones/<id>/ and POST /sync/push call milestone.get_incomplete_prerequisites() and return HTTP 400 if any predecessor is <100%.
- **Zero Premature Drawdowns:** Protects developers and lenders from paying for superstructure or finishes before foundational integrity is 100% certified.



GOVERNANCE GUARANTEE Stage-gates turn construction contracts into executable code—eliminating subjective certification bias.

05 • QUALITY, STRUCTURAL TESTING & HSE

Field Snagging, Concrete Cube Lifecycle & Safety Escalation

Capturing structural quality tests and engineering defects at the source with live Kanban synchronization.

Mobile Snag & Defect Ticketing

Resident Engineers log structural and finishing defects directly from the floor plate.

- **Severity & Category:** Urgent / High / Normal routing across Structural, MEP, and Finishes.
- **Geotagged Photo Proof:** Mandatory camera capture with hardware GPS coordinates.
- **Live Office Kanban:** Syncs immediately to the Web ERP oversight board for contractor resolution.
- **Closure Verification:** Requires before-and-after photographic sign-off.

Concrete Cube Curing Scheduler

Automated structural pour compliance aligned with BS 8110 / Eurocode 2 standards.

- **Batch & Slump Intake:** Logs concrete grade (e.g. C30/37), mixer truck, and slump test results.
- **Automated 7 / 14 / 28-Day Alerts:** Automatically schedules mandatory lab crushing tests.
- **Pass/Fail Thresholds:** Highlights compressive strength shortfalls (MPa) before slab loading.
- **Permanent Audit Archive:** Lab certificates attached directly to the structural milestone.

HSE & Regulatory Compliance

Immediate escalation protocol for site safety inspections, near-misses, and NCA/DOSHS compliance.

- **PPE & Scaffold Audits:** Daily digital checklists completed by the Clerk of Works.
- **Incident Root-Cause Tracking:** Severity grading and Corrective Action Plan (CAP) workflows.
- **Statutory Readiness:** Maintains full National Construction Authority (NCA) inspection logs.
- **Zero-Delay Escalation:** High-severity incidents trigger instant executive alerts.

GOVERNANCE GUARANTEE **Quality assurance records and concrete test certificates form an immutable structural passport for handover.**

06 • COMMERCIAL & FINANCIAL CONTROL

Earned Value S-Curve & Automated Mini-BoQ Valuations (IPCs)

Converting verified field progress into audit-ready Interim Payment Certificates in minutes.

QS 1. Mini-BoQ Line-Item Valuation

Quantity Surveyors inspect the exact BoQ sub-elements linked to each milestone, calculating certified KES earned value with zero spreadsheet formulas.

AI 2. AI Progress & Variance Audit

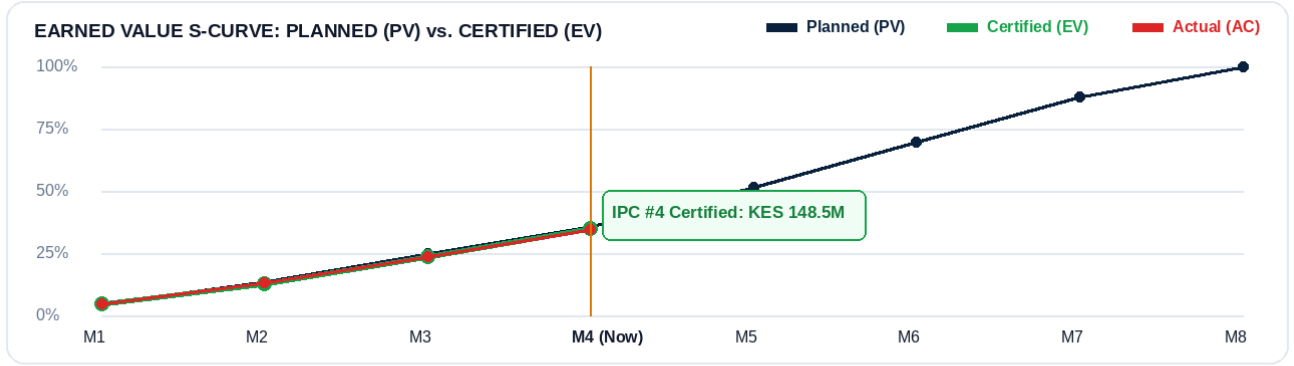
Automated variance engine compares Planned Value (PV), Earned Value (EV), and Actual Cost (AC), flagging schedule slippage and material anomalies.

IPC 3. Escrow & Retention Drawdowns

Generates bank-ready PDF Interim Payment Certificates with automatic 10% retention deductions and cryptographic links to supporting site photos.

Center Estates Web ERP • Karam Apartments — Lang'ata (PRJ-KRM-01) • Commercial & IPC Valuation

CONTRACT SUM (BoQ)	CERTIFIED TO DATE (IPC #4)	RETENTION HELD (10%)	MATERIAL VARIANCE
KES 420.0M	KES 148.5M	KES 14.85M	-0.4% (SAVING)
Approved Baseline	35.4% Earned Value	Escrow Protected	BoQ Shield Active



BoQ CODE	ELEMENT DESCRIPTION	BUDGET (KES)	VERIFIED %	CERTIFIED IPC	AUDIT STATUS
BOQ-01.01	Substructure & Raft Foundation	84,000,000	100.0%	84,000,000	✓ SHA-256 Sealed
BOQ-02.04	Superstructure RC Frame (Fl 1-3)	112,000,000	57.6%	64,500,000	✓ GPS + Photo Verified
BOQ-03.01	Roofing & Structural Waterproofing	48,000,000	0.0%	0	🔒 Stage-Gate Locked

GOVERNANCE GUARANTEE Every shilling certified on an Interim Payment Certificate is backed by geotagged photos and sealed gate deliveries.

07 • ANTI-FRAUD ARCHITECTURE

Offline Resilience & Cryptographic Non-Repudiation

Purpose-built for deep basements and remote sites without sacrificing bank-grade security.

01 Offline SQLite Engine

Deep excavations and basements rarely have reliable 4G.

- **Local Persistence:** All BoQ items, milestones, and daily logs cached in SQLite.
- **Draft Recovery:** useFormDraft hook prevents data loss if battery dies mid-inspection.
- **Auto Delta-Sync:** Queued payloads sync atomically once connectivity returns.

02 Biometric Hardware Keys

Eliminates shared passwords and unauthorized proxy sign-offs.

- **Ed25519 Keypair:** Generated inside the mobile device's secure hardware keystore.
- **Biometric Prompt:** Fingerprint / FaceID required to sign milestone submissions.
- **Non-Repudiation:** Cryptographic proof of the exact engineer who certified the work.

03 GPS Geofence Guard

Prevents off-site photo uploads and Recycled Photo fraud.

- **Perimeter Lock:** Compares hardware GPS coordinates against project polygon.
- **Distance Telemetry:** Records exact meter offset and GPS accuracy radius.
- **Fraud Flagging:** Out-of-bounds submissions are automatically flagged for audit.

04 SHA-256 Evidence Seals

Tamper-evident cryptographic sealing at the moment of capture.

- **Composite Hash:** Computed over milestone ID, progress %, notes, GPS, and photo bytes.
- **Server Verification:** Re-verified upon ingestion by api.centerestates.com.
- **Audit Certificate:** Displayed on Web ERP alongside the green Verified badge.

GOVERNANCE GUARANTEE **Security is embedded directly into the mobile data capture layer—making fraudulent progress claims mathematically impossible.**

08 • LIVE DEMONSTRATION SCRIPT

Recommended 5-Step Live Demonstration Flow (8 Minutes)

A high-impact, cohesive walkthrough using Karam Apartments — Lang'ata (PRJ-KRM-01).

STEP 01	**Office Baseline Setup (Web ERP)**	Open Karam Apartments (PRJ-KRM-01). Show the CPM Work Plan Gantt and approved BoQ allowances.	1.5 MIN
STEP 02	**Site Gate Delivery (Mobile App)**	Open Daily Log -> Material Delivery. Pick 'BOQ-02.04 Cement', tap 'Bamburi', show Planned Allowance Shield.	2.0 MIN
STEP 03	**Stage-Gate Lockout (Mobile App)**	Open 'Superstructure Frame'. Highlight the red '🔒 Contractual Stage-Gate Locked' banner blocking claims.	1.5 MIN
STEP 04	**Live Defect Sync (Mobile -> Web)**	Capture a rebar spacing snag on Mobile. Switch to Web ERP to show immediate appearance on the Kanban board.	1.5 MIN
STEP 05	**Mini-BoQ & IPC Valuation (Web)**	Open Milestones -> Mini-BoQ Valuation. Inspect certified KES value, retention deduction, and AI Progress Audit.	1.5 MIN

GOVERNANCE GUARANTEE **Keep the entire demonstration anchored inside Karam Apartments (PRJ-KRM-01) to prove end-to-end data continuity.**

09 • BUSINESS CASE & ROI

Measurable Financial Impact Across Every Project Stakeholder

Why institutional developers, consultants, and project financiers standardize on Center Estates.

DEV Real Estate Developers

Protects equity capital and accelerates project delivery.

- **15%–20% Cost Protection:** Eliminates phantom deliveries and unapproved BoQ overages.
- **Real-Time Visibility:** Live executive S-Curve across all active development sites.
- **Seamless Handover:** Completed units transition directly into Center Estates Sales & Lettings.

QS Project Managers & QSs

Replaces manual spreadsheet reconciliation with automated governance.

- **80% Faster IPC Valuations:** Mini-BoQ calculations generated in minutes instead of days.
- **Standardized Site Records:** Automated concrete cube schedules and daily weather/labour logs.
- **Dispute Defense:** Time-stamped, geotagged photographic proof resolves contractor claims.

FIN Banks & Project Financiers

De-risks construction debt drawdowns with cryptographic proof.

- **Verified Drawdowns:** Every tranche disbursement tied to 100% stage-gated milestones.
- **Anti-Collusion Controls:** Ed25519 biometric signatures + GPS geofence validation.
- **ESG & Statutory Audit:** Complete NCA, DOSHS, and structural quality compliance trail.

GOVERNANCE GUARANTEE Center Estates transforms construction monitoring from a passive reporting chore into an active financial shield.

10 • SUMMARY & LIVE Q&A

One Operating Record: From Groundbreaking to Estate Handover

The mobile app is live, the backend is synchronized, and Karam Apartments is ready for inspection.

GATE INTAKE (BoQ) → FIELD VERIFICATION → STAGE-GATE GUARD → IPC CERTIFICATION → ESTATE HANDOVER

Ready for Live Walkthrough & Stakeholder Scenarios

Live Demonstration Environment

- **Web ERP Command Center:** https://centerestates.com
- **Presentation & Resource Hub:** https://centerestates.com/presentations/
- **Featured Project:** Karam Apartments — Lang'ata (PRJ-KRM-01)
- **Automated Test Suite:** 100% Pass Rate (135 Web + 73 Mobile Tests)

Platform & Engineering Governance

- **Suite:** Center Estates Enterprise Construction & Field Mobility
- **Architecture:** Next.js 14 Web • Expo React Native • Django REST API
- **Security:** Ed25519 Biometrics • SHA-256 Seals • SQLite Offline Sync
- **Organization:** Center Estates Enterprise Engineering

GOVERNANCE GUARANTEE

Thank you. Let us open Karam Apartments (PRJ-KRM-01) for the live demonstration.